

```
USE ROLE ACCOUNTADMIN;  
USE WAREHOUSE COMPUTE_WH;  
USE DATABASE SNOWFLAKE_FUNDAMENTALS;  
USE SCHEMA DAY2;
```

```
/*=====Analytical Functions=====
```

SQL analytical functions, also known as window functions, are a powerful set of functions in SQL used to perform calculations across a set of table rows that are related to the current row. Unlike aggregate functions (like SUM, AVG, COUNT), analytical functions do not group rows into a single output row—instead, they return a value for each row while considering a window (a set of rows related to the current one).

```
=====*/
```

```
/*=====RANK FUNCTION=====
```

Returns the rank of a value within an ordered group of values.

The rank value starts at 1 and continues up sequentially.

If two values are the same, they have the same rank.

```
=====*/
```

```
-- Create table and load data.
```

```
create or replace table corn_production (  
  farmer_ID INTEGER,  
  state varchar,  
  bushels float  
);
```

```
insert into corn_production (farmer_ID, state, bushels) values
```

```
  (1, 'Iowa', 100),  
  (2, 'Iowa', 110),  
  (3, 'Kansas', 120),  
  (4, 'Kansas', 130),  
  (4, 'Iowa', 110);
```

```
--Show farmers' corn production in descending order, along with the rank of each  
individual farmer's production:
```

```
SELECT  
  state,  
  bushels,  
  RANK() OVER (ORDER BY bushels DESC)  
FROM corn_production;
```

/*=====DENSE RANK FUNCTION=====

The query and output below show how tie values are handled by the RANK and DENSE_RANK functions.

Note that for DENSE_RANK, the ranks are 1, 2, 3, 3, 4.

Unlike with the output from the RANK function, the rank 4 is not skipped because there was a tie for rank 3.

=====*/

```
SELECT
state,
bushels,
RANK() OVER (ORDER BY bushels DESC),
DENSE_RANK() OVER (ORDER BY bushels DESC)
FROM corn_production;
```